ENGINEERING BRIEF · AGENT E · NEW ONBOARDING

Roomio — Lighting & Time-of-Day

Welcome aboard, Agent E. Your domain is light: give every room sensible default lights the user can replace, add layered ambient/task/accent lighting, and build a toggleable time-bar with a sun that casts real, moving shadows driven by time and a rotatable north indicator. This is what makes Roomio look like a room, not a 3D model.

SUN + SHADOWS | TIME-OF-DAY BAR | LAYERED ROOM LIGHTS

Repo: `~/Desktop/personal/roomio` · **Your folder:** `/lighting`
Coordination: `/shared/*.json` + `roomio.txt` + `/shared/LEARNINGS.md`
Date: June 29, 2026 · **Coordinator:** Agent D merges your work

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1. Onboarding: The Team & Your Place

> You are **Agent E**, joining a Roomio build already in motion. First action: read `roomio.txt` end-to-end and skim `/shared/LEARNINGS.md` to absorb project history and hard-won codebase facts before writing code. Then post an onboarding entry.

Agent	Domain	Owns
A	Room interior	Shape wizard, furnishing, archetype library, persona presets, design-suggestion engine.
B	Detection + camera	Furniture-detection pipeline; virtual-camera flythrough.
C	The house	Multiple rooms + connectors (doorways, openings, archways).
D	Coordinator	Merges everyone to a runnable main, verifies contracts, curates `LEARNINGS.md`. **D merges your work too.**
E (you)	Lighting	Default room lights, layered lighting, sun + shadows, time-of-day bar + north control.

HOW YOU FIT

Lighting touches everyone's work: it illuminates A's furniture, lives in C's multi-room house (every room needs light), and the camera (B) records whatever lighting you set. So coordinate tightly via `/shared` and `LEARNINGS.md`, and let Agent D catch integration issues. Your lights are scene-level; design them to apply per-room across C's house, not just one room.

2. Mission & Three Pillars

Lighting is what separates "looks like a 3D model" from "looks like a room." Three pillars, buildable in order:

#	Pillar	What it delivers
P1	Default room lights	Every room gets a sensible default ceiling light automatically, plus layered ambient/task/accent options. User can replace or adjust any of it.
P2	Sun & real shadows	A directional "sun" that casts moving, realistic shadows through windows and across furniture.
P3	Time bar + north	A toggleable top bar mimicking time-of-day; sun position follows the bar; a north indicator that can be rotated and reversed, changing where light comes from.

TIE INTO A'S EXISTING RULEBOOK

Agent A's design-suggestion engine already preaches *layered lighting* (ambient + task + accent) and flags rooms with only one harsh overhead light. Your default-lights system should satisfy those rules out of the box and expose the layers A's suggestions point users toward. Coordinate so a room you light passes A's lighting suggestions.

3. Pillar 1 — Default Room Lights (layered)

Every room should be lit sensibly the moment it exists — the user shouldn't face a dark box. Provide defaults, make them replaceable.

The layers (the documented three-layer model)

- **Ambient** — soft global fill so nothing is pitch black. A low-intensity `AmbientLight` or, better, a `HemisphereLight` (sky + ground tint) for more natural fill.
- **Task** — functional light where activity happens: a ceiling fixture per room (the default everyone gets), plus task lights (desk/reading) where relevant.
- **Accent** — optional highlights (wall wash, under-cabinet, art) for depth and mood.

Defaults & replaceability

- On room creation, auto-add one ceiling light (task) + ambient fill. This is the "every room gets basics" requirement.
- Each default is a normal, editable light object — user can move it, change color/intensity/warmth, delete it, or swap it for a fixture from a light library (pendant, floor lamp, sconce).
- Warmth matters: offer warm (~2700K-ish) vs cool (~5000K-ish) presets — this single control hugely changes how "real" a room feels.

PERFORMANCE DISCIPLINE

Shadowed real-time lights are expensive. Cap shadow-casting lights to a few; the sun should be the primary shadow caster, with most room lights either non-shadowing or softly shadowing. Disable `castShadow` on lights that don't need it. For a multi-room house, only light/shadow the rooms in view where feasible.

4. Pillar 2 — The Sun & Real Shadows

The sun is a single **DirectionalLight** — its rays are parallel, exactly like real sunlight from an effectively infinite distance. Its *direction* is what matters (position → target), and that direction is what the time bar will drive.

```
// Sun = directional light with shadows
const sun = new THREE.DirectionalLight(0xffffff, 1.0);
sun.castShadow = true;
sun.shadow.mapSize.set(2048, 2048);     // resolution; raise for sharper shadows
// Orthographic shadow frustum MUST cover the whole house:
const d = 20;                           // half-extent ~ house size
sun.shadow.camera.left = -d; sun.shadow.camera.right = d;
sun.shadow.camera.top = d;  sun.shadow.camera.bottom = -d;
sun.shadow.camera.near = 0.5; sun.shadow.camera.far = 100;
sun.shadow.bias = -0.0005;              // fight shadow acne (see §6)
scene.add(sun); scene.add(sun.target);   // target must be in the scene
```

Critical setup notes

- **Direction, not rotation.** A DirectionalLight aims from its `position` toward its `target`; setting its rotation does nothing. The time bar moves the `position` around the scene; the target stays at house center.
- **Shadow camera is orthographic** (parallel rays). Its bounds (`left/right/top/bottom`) must enclose the whole house or shadows clip; sized too large and they get coarse. Tune to house extent.
- **Renderer + meshes.** Enable `renderer.shadowMap.enabled = true` ; set `castShadow` on furniture/walls and `receiveShadow` on floors/walls. Activate on as few objects as needed for performance.
- **Soft shadows.** Use `PCFSoftShadowMap` for antialiased, realistic edges (vs hard pixelated BasicShadowMap).
- **Sky + ambient.** Pair the sun with a hemisphere/ambient fill so shadowed areas aren't pure black, and optionally a sky background/gradient that shifts with time.

5. Pillar 3 — Time Bar + North Indicator

> A toggleable top bar represents time of day. As the user scrubs it, the sun arcs across the sky (dawn → noon → dusk), shadows sweep accordingly, and color temperature shifts (warm low sun, neutral midday). A **north indicator** sets which way is north; rotating or reversing it changes where the sun comes from. The whole bar/sign toggles on/off to hide it.

Driving the sun from the bar (recommended approach)

You do NOT need real geolocation for a design tool. Map the bar directly: bar position (0–24h) → a sun **elevation** (low at dawn/dusk, high at noon) and **azimuth** (east→south→west across the day). Then offset the azimuth by the user's **north rotation**. Convert (azimuth, elevation) to a 3D position on a dome around the house and place the sun there.

```
// time (0..1 over the day) + northOffset (radians) → sun position
const elev = Math.sin(time * Math.PI) * maxElev;   // 0 at dawn/dusk, peak at noon
const az   = (time * Math.PI) - Math.PI/2 + northOffset; // E→W sweep, rotated by north
const r = 50;
sun.position.set(
  r * Math.cos(elev) * Math.sin(az),
  r * Math.sin(elev),
  r * Math.cos(elev) * Math.cos(az));
sun.target.position.set(0,0,0); sun.target.updateMatrixWorld();
sun.intensity = Math.max(0, Math.sin(time * Math.PI)); // dim near horizon/night
```

North indicator controls

- A compass/north sign at the top, **toggleable on/off** (hide it when not needed).
- **Rotate** north (drag/step) — changes `northOffset`, so the sun and shadows swing to match the building's real orientation.
- **Reverse** north (flip 180°) — a quick way to send light from the opposite side.
- The bar and the north sign are independent toggles; default the bar to a pleasant mid-morning/afternoon angle so rooms look good immediately.

Optional: astronomical accuracy

If you later want *real* sun positions for a real location/date, the **SunCalc** library (BSD-licensed, tiny) returns azimuth + altitude from time + lat/long. For a design tool the direct bar→arc mapping above is simpler and gives the user full control; treat SunCalc as an optional "use my real location" mode, not the default.

6. Shadow Quality: the gotchas

Shadows are where this feature looks great or looks broken. The two classic failure modes and their fixes:

Problem	Looks like	Fix
Shadow acne	Black speckles/moiré on lit surfaces (surface shadows itself)	Apply a small negative `shadow.bias` (e.g. -0.0005); prefer `normalBias` for curved/ angled surfaces.
Peter-panning	Shadow detaches from its caster, floats away	Bias too large — reduce it. Balance against acne.
Coarse/blocky shadows	Pixelated, low-res shadow edges	Raise `shadow.mapSize` (2048+) and/or tighten the orthographic shadow-camera bounds to the actual scene.
Clipped shadows	Shadows cut off at room edges	Shadow-camera frustum too small — widen `left/right/top/bottom` and `far` to enclose the house.
Hard jagged edges	Aliased, sharp shadow borders	Use `PCFSoftShadowMap`; consider VSM for softest (needs opaque receivers).

TUNING IS SCENE-SCALE DEPENDENT

The "right" bias and shadow-camera size depend on Roomio's world units and house size. There's no universal value — tune against an actual furnished room, and record the values that work in `LEARNINGS.md` so other agents (and future you) don't re-derive them.

INTENSITY UNITS NOTE

Modern Three.js (r155+) uses physically-based light units by default (legacy lights off). If intensities look wrong vs an old tutorial, that's why — tune to the current renderer rather than copying old intensity numbers. Record working values in LEARNINGS.md.

7. Stack, Schema & Coordination

Stack

Need	Tool	Why
Sun	THREE.DirectionalLight	Parallel rays = real sunlight; casts orthographic shadows.
Ambient fill	HemisphereLight / AmbientLight	Sky+ground tint or flat fill so shadows aren't black.
Room/task/accent	PointLight / SpotLight / RectAreaLight	Bulbs, cones, soft panels. RectAreaLight needs its uniforms lib + PBR materials.
Soft shadows	PCFSoftShadowMap	Antialiased realistic edges.
Real sun (optional)	SunCalc (BSD)	azimuth+altitude from time+location, if you add a "real location" mode.
If R3F	drei (<Environment>, <SoftShadows>)	Match Agent A's setup; drei has lighting/shadow helpers. Confirm via LEARNINGS.md.

Lighting schema (publish to /shared)

```
{
  "time_of_day": 0.55,            // 0..1 across the day
  "north_offset_deg": 30,         // rotatable; +180 = reversed
  "bar_visible": true, "north_visible": true,
  "sun": { "enabled": true, "max_elevation_deg": 60, "warmth_shift": true },
  "rooms": {
    "r_living": {
      "lights": [
        { "id":"ceil_1","type":"ceiling","layer":"task","color":"#fff3e0",
          "intensity":1.0,"warmth":"warm","cast_shadow":true,"default":true,"pos":[..]}
      ]
    }
  }
}
```

Coordination (same protocol as A/B/C)

- Code in `/lighting`; only other writable places are `/shared` and `roomio.txt` + `LEARNINGS.md` notes. Commit only your own paths — **Agent D owns merges**.
- Read `roomio.txt` + `LEARNINGS.md` at the start of every cycle (esp. renderer setup: plain Three.js vs R3F, and the scene/house schemas A and C publish). Watch for `[USER -> AGENT-E]` messages as top priority.
- Log INTENT/DECISIONS; post `REQUEST -> AGENT-A` (light up its furniture, satisfy its lighting suggestions), `REQUEST -> AGENT-C` (light per-room across the house; need window positions for sun shafts), `REQUEST -> AGENT-B` (camera should record your lighting state). Treat `ISSUE -> AGENT-E` from D as high priority.
- When you find tuned values (bias, shadow size, intensities), record them in `LEARNINGS.md`.

8. Functional Requirements & Acceptance

ID	Requirement	Pillar
L-1	Every room auto-gets a default ceiling light + ambient fill on creation.	P1
L-2	Default lights are editable: move, recolor, set intensity/warmth, delete, or swap for a fixture.	P1
L-3	Layered lighting available (ambient + task + accent); satisfies A's lighting suggestions.	P1
L-4	Warm/cool color-temperature control.	P1
L-5	A directional sun casts realistic soft shadows from furniture/walls onto floors.	P2
L-6	Shadows are clean (no acne/peter-panning), cover the whole house, soft-edged.	P2/P6
L-7	A toggleable time bar; scrubbing it arcs the sun and sweeps shadows dawn→noon→dusk.	P3
L-8	Sun color/intensity shifts with time (warm low sun, dim near night).	P3
L-9	A toggleable north indicator; rotating it swings the sun direction; reverse flips 180°.	P3
L-10	Bar and north sign hide/show independently; sensible default angle on load.	P3
L-11	Lighting applies per-room across C's multi-room house; performant (few shadow casters).	All
L-12	Code in /lighting; coordinate via /shared + roomio.txt + LEARNINGS.md; D merges; no edits to others' source.	Coord

Acceptance test (the "done" bar)

Open a furnished room: it's already lit (no dark box) with an editable default ceiling light; switch a light to warm and the room feels cozier. Turn on the sun: furniture casts clean soft shadows on the floor. Show the time bar and scrub it: the sun arcs across the sky, shadows sweep across the room, and the light warms at the low-angle ends. Show the north sign and rotate it: the sun direction swings to match; reverse it and light comes from the opposite side. Hide both controls and the scene still renders with the chosen lighting. In a multi-room house, each room is lit and shadows render without tanking framerate.

9. Milestones

Milestone	Deliverable	Definition of done
E0	Read roomio.txt + LEARNINGS.md; scaffold /lighting; publish lighting schema to /shared; post onboarding entry; confirm renderer (Three vs R3F) with A	Onboarded; schema in /shared; A/C pinged.
E1	P1: default ceiling + ambient per room; editable lights; warm/cool	No room is a dark box; lights editable.
E2	Layered ambient/task/accent; satisfies A's lighting suggestions	A's "needs layered lighting" rule passes on a default room.
E3	P2: directional sun + soft shadows; shadow-camera + bias tuned	Clean soft shadows across a furnished room; values logged to LEARNINGS.md.
E4	P3: time bar drives sun arc + intensity/warmth shift	Scrubbing sweeps shadows and changes light color.
E5	P3: north indicator rotate/reverse; bar + sign toggles	Rotating north swings the sun; both controls hide/show.
E6	Multi-room application + performance pass + polish	House-wide lighting; framerate holds; cleanup.

One-line summary. In `/lighting` : auto-light every room with editable layered defaults (ambient/task/accent, warm/cool), add a DirectionalLight sun with clean soft shadows (PCFSoft, tuned bias + orthographic frustum), and a toggleable time bar + rotatable/reversible north indicator that drive the sun's arc, shadow sweep, and color across the day. Coordinate through /shared + roomio.txt + LEARNINGS.md; Agent D merges. No NeRF, no real geo required — the bar drives the sun directly.

End of brief · Roomio Lighting & Time-of-Day (Agent E) · Tune shadow values against a real scene and record them in LEARNINGS.md.